

N HECATE STRAIT BUOY 46183, Canada

WMO: 996460

Lat: 53.620N

Lon: 131.100W

Elev: 10

StdP: 14.69

Time Zone: -9.00 (W09)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.1	31.9	N/A	N/A	N/A	N/A	N/A	N/A	44.0	43.9	41.1	44.3	16.2	20	0.839

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.9	61.1	N/A	60.0	N/A	59.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	9.4	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 41.2	36.9	33.2	DB	24.3	63.4	6.6	2.8	19.5	65.4	15.6	67.1	11.9	68.6	7.1	70.7	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	42.8	42.0	42.0	44.5	48.4	52.5	55.8	57.1	54.6	50.5	45.4	42.9
	DBStd	6.29	4.67	3.77	3.88	2.36	2.36	2.44	2.15	1.86	1.77	2.48	3.81	4.22
	HDD50	1306	224	224	248	165	59	4	0	0	0	23	138	221
	HDD65	6112	688	644	713	614	513	374	286	245	312	450	587	686
	CDD50	668	0	0	0	0	11	80	179	220	138	38	2	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.3	19.6	18.1	18.7	16.6	13.7	13.8	12.7	11.8	14.3	17.1	19.3	20.1
Precipitation	PrecAvg	70.0	8.2	6.0	5.8	4.4	3.1	2.5	2.3	3.1	4.7	9.3	9.2	9.6
	PrecMax	83.5	13.4	11.5	8.5	7.0	6.4	4.5	4.9	7.3	9.8	16.6	17.0	19.7
	PrecMin	58.3	4.7	2.2	3.5	2.5	1.1	1.1	0.9	0.9	1.1	4.4	5.3	4.4
	PrecStd	7.3	2.2	2.4	1.5	1.1	1.4	0.9	1.0	1.6	1.9	2.9	2.5	3.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.7	50.1	50.0	52.3	55.4	61.5	61.8	63.1	60.2	56.2	53.0	51.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	49.5	48.5	48.6	49.8	53.4	58.7	60.3	61.6	58.9	54.9	51.3	49.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	48.5	47.2	47.5	48.5	52.2	57.2	59.4	60.5	57.9	54.1	50.7	48.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	47.6	46.4	46.5	47.6	51.3	55.9	58.7	59.7	57.2	53.4	49.9	48.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	4.0	3.8	3.9	3.3	2.9	2.8	2.9	2.9	2.7	3.1	4.1	4.0
		MCDBR	3.9	3.9	4.0	3.8	3.8	3.9	3.7	4.0	3.3	3.4	4.4	4.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.312	0.320	0.345	0.380	0.377	0.373	0.371	0.370	0.367	0.346	0.323	0.305	
	taud	2.405	2.491	2.434	2.326	2.399	2.450	2.496	2.515	2.484	2.445	2.339	2.326	
	Ebn at Noon	211	251	266	269	277	278	277	271	256	235	199	184	
	Edh at Noon	17	22	28	36	35	34	31	29	27	23	18	16	
All-Sky Solar Radiation	RadAvg	176	441	797	1253	1581	1653	1653	1550	1321	890	499	233	143
	RadStd	22	70	90	69	136	118	118	163	104	93	63	27	21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page